Impact of religious behavior: Secondary analysis among SDA sample

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1 Introduction

A cross-sectional online survey was conducted to evaluate spirituality, religious behavior and belief, religious coping, mental health, and perceived stress of healthy adults (aged 18 years or above) in Hong Kong from August to October 2020. Respondents were recruited by convenience and snowball sampling and asked to complete a structured questionnaire, including Functional Assessment of Chronic Illness Therapy-Spiritual Well-Being (FACIT-Sp-12), Impact of Event Scale – Revised (IES - R), Brief Religious Coping Activity Scales (Brief RCOPE), Hospital Anxiety and Depression Scale (HADS), the self-developed instruments about religious behavior and belief. As an incentive to participate in the study, participants were invited to enter names and contact information into a lucky draw. Ten participants would be selected to win a range of HK$ 200-500 (US$25-65) supermarket coupons.

A total of 418 responses were received for the study. Excluding those with duplicated submissions, there were 360 valid responses. On those valid responses, only 112 participants self-reported that they were Seventh-day Adventist (31.1%). Of the 112 respondents included in this secondary analysis, 11% and 14% were borderline abnormal for anxiety and depression respectively, while 6.2% and 8.9% were abnormal for anxiety and depression respectively. Multiple linear regression analysis found that attending religious meetings regularly was associated with increased faith and positive religious coping, and reduced depression and intrusion perceived stress. While people with regular personal praying life also had increased peace, faith, and positive religious coping. However, there were no associations between studying religious books/materials regularly and spirituality, religious coping, mental health, and perceived stress.

於 2020 年 8 月至 10 月進行了一項橫斷面在線調查，以評估香港健康成年人 (18 歲或以上) 的靈性、宗教行為和信念、宗教應對、心理健康和自我認知壓力。透過滾雪球式取樣方式招募參加者，參加者要完成問卷，包括靈性 (FACIT-Sp-12)、自我認知壓力 (IES-R)、宗教應對 (Brief RCOPE)，焦慮與抑鬱量 (HADS)，及其他關於宗教行為和信念問卷。作為參與研究的獎勵，完成問卷的參與者會被邀請進行抽獎，並從中選出十名參與者，贏取價值 200-500 港元 (25-65 美元) 的超市優惠券。

本研究共收到 418 份參加者的回應，剔除了重複的回應後，共有 360 名參加者，只有 112 名參與者，自稱他們是基督復臨安息日會的信徒 (31.1%)。在這 112 名參加者中當中，分別有 11% 和 14% 是邊緣焦慮和邊緣抑鬱；而焦慮和抑鬱則分別有 6.2% 和 8.9%。多元線性回歸分析發現，定期參加宗教聚會與增加信心和積極的宗教應對有關，並能減少抑鬱和侵入性的自我認知壓力。此外，有規律的個人祈禱生活的人也令人更平安、更有信心、及更多使用正面的宗教應對方法。然而，閱讀宗教書籍與靈性、宗教應對、心理健康和自我認知壓力之間沒有關係。

2 Profile of HKMC SDA

- Characteristic of church member attending religious meeting regularly
  1. higher education level
  2. higher peace
  3. higher faith
  4. more personal pray
  5. more studying religious books/materials
  6. lower intrusion perceived stress
  7. lower hyperarousal perceived stress
  8. more positive religious coping
  9. lower anxiety
  10. lower depression
- On those church members attending religious meetings every week (46%), only 56% had personal praying life every day and 27% studied religious books/materials every day. Besides, on those church members attending religious meetings more than 1 time per week (34%), only 82% had personal praying life every day and 68% studied religious books/materials every day. Therefore, even though people attend religious meetings regularly, they may not have personal praying life and study religious books/materials every day.

- 恆常出席宗教聚會的教友的特徵
  1. 教育水平較高
  2. 內心更平安
  3. 內心更有信心
  4. 更個人的祈禱
  5. 較多閱讀宗教書籍
  6. 入侵感知壓力較低
  7. 過度喚醒感知壓力較低
  8. 較多運用正面的宗教應對方法來處理壓力
  9. 焦慮較低
  10. 抑鬱較低
- 在那些每週出席宗教聚會的教友 (46%)，只有 56% 每天都有個人的祈禱生活和 27% 每天閱讀宗教書籍。此外，對於那些每週出席超過一次宗教聚會的教友 (34%)，只有 82% 每天都有個人的祈禱生活，68% 每天閱讀宗教書籍。因此，即使教友恆常出席宗教聚會，他們也可能沒有每天的個人祈禱生活和閱讀宗教書籍。

Table 1: Demographic characteristic of the respondents and comparison among religious meeting attendance

Characteristic	Overall, N = 112[1]	< 1/week, N = 22[1]	1/week, N = 52[1]	> 1/week, N = 38[1]	p[2]
Age					0.287
18-24 years	12 (11%)	3 (14%)	5 (9.6%)	4 (11%)	
25-44 years	48 (43%)	5 (23%)	24 (46%)	19 (50%)	
45 years or above	52 (46%)	14 (64%)	23 (44%)	15 (39%)	
Gender					0.915
Female	82 (73%)	16 (73%)	39 (75%)	27 (71%)	
Male	30 (27%)	6 (27%)	13 (25%)	11 (29%)	
Education					**0.022**
Secondary or below	19 (17%)	6 (27%)	6 (12%)	7 (18%)	
Post-secondary	16 (14%)	5 (23%)	10 (19%)	1 (2.6%)	
Bachelor	44 (39%)	8 (36%)	23 (44%)	13 (34%)	
Master or above	33 (29%)	3 (14%)	13 (25%)	17 (45%)	
Employment status					0.364
Full time	69 (62%)	10 (45%)	34 (65%)	25 (66%)	
Self employed	7 (6.2%)	2 (9.1%)	1 (1.9%)	4 (11%)	
Part time	16 (14%)	6 (27%)	7 (13%)	3 (7.9%)	
Unemployed	5 (4.5%)	1 (4.5%)	3 (5.8%)	1 (2.6%)	
Retired	8 (7.1%)	1 (4.5%)	5 (9.6%)	2 (5.3%)	
Student	7 (6.2%)	2 (9.1%)	2 (3.8%)	3 (7.9%)	
Salary					0.107
No income	17 (16%)	6 (30%)	7 (14%)	4 (11%)	
13,100 or below	16 (15%)	4 (20%)	8 (16%)	4 (11%)	
13,101-18,200	21 (20%)	3 (15%)	11 (22%)	7 (19%)	
18,201-28,200	21 (20%)	2 (10%)	9 (18%)	10 (27%)	
28,201-44,600	16 (15%)	4 (20%)	3 (6.1%)	9 (24%)	
44,601 or above	15 (14%)	1 (5.0%)	11 (22%)	3 (8.1%)	

[1]n (%)

[2]Fisher's exact test; Pearson's Chi-squared test; Fisher's Exact Test for Count Data with simulated p-value
(based on 2000 replicates)

Table 2: Spirituality characteristic of the respondents and comparison among religious meeting attendance

Characteristic	Overall, N = 112[1]	< 1/week, N = 22[1]	1/week, N = 52[1]	> 1/week, N = 38[1]	p[2]
Meaning	9.9 (1.7)	9.3 (1.9)	9.8 (1.7)	10.4 (1.7)	0.058
Peace	10.9 (3.2)	8.9 (3.3)	10.9 (3.2)	11.9 (2.6)	**0.001**
Faith	11.5 (3.4)	9.5 (3.9)	11.7 (3.1)	12.6 (3.0)	**0.003**

[1]Mean (SD)

[2]One-way ANOVA

Table 3: Religious behaviour and belief characteristic of the respondents and comparison among religious meeting attendance

Characteristic	Overall, N = 112[1]	< 1/week, N = 22[1]	1/week, N = 52[1]	> 1/week, N = 38[1]	p[2]
Self-perceived devotion	7.1 (1.9)	5.4 (2.5)	7.3 (1.4)	7.8 (1.5)	**<0.001**
Personal pray					**<0.001**
< 1/day	45 (40%)	15 (68%)	23 (44%)	7 (18%)	
>= 1/day	67 (60%)	7 (32%)	29 (56%)	31 (82%)	
Studying religious books/materials					**<0.001**
< 1/day	70 (62%)	20 (91%)	38 (73%)	12 (32%)	
>= 1/day	42 (38%)	2 (9.1%)	14 (27%)	26 (68%)	

[1]Mean (SD); n (%)

[2]One-way ANOVA; Pearson's Chi-squared test

Table 4: Religious coping characteristic of the respondents and comparison among religious meeting attendance

Characteristic	Overall, N = 112[1]	< 1/week, N = 22[1]	1/week, N = 52[1]	> 1/week, N = 38[1]	p[2]
Positive	20.2 (5.0)	17.0 (4.3)	20.5 (5.2)	21.8 (4.5)	**0.001**
Negative	9.4 (2.2)	9.8 (3.1)	9.3 (2.0)	9.3 (2.0)	0.689

[1]Mean (SD)

[2]One-way ANOVA

Table 5: Perceived stress characteristic of the respondents and comparison among religious meeting attendance

Characteristic	Overall, N = 112[1]	< 1/week, N = 22[1]	1/week, N = 52[1]	> 1/week, N = 38[1]	p[2]
Intrusion	5.4 (4.0)	8.1 (4.7)	4.9 (2.9)	4.5 (4.2)	**<0.001**
Avoidance	5.9 (5.2)	7.8 (5.3)	5.1 (4.9)	6.0 (5.4)	0.127
Hyperarousal	3.8 (3.1)	5.5 (4.3)	3.4 (2.6)	3.4 (2.7)	**0.022**

[1]Mean (SD)

[2]One-way ANOVA

Table 6: Mental health characteristic of the respondents and comparison among religious meeting attendance

Characteristic	Overall, N = 112[1]	< 1/week, N = 22[1]	1/week, N = 52[1]	> 1/week, N = 38[1]	p[2]
Anxiety	4.7 (3.6)	6.8 (4.5)	4.3 (3.2)	3.8 (3.1)	**0.005**
Anxiety category					**0.022**
Abnormal	7 (6.2%)	5 (23%)	2 (3.8%)	0 (0%)	
Borderline abnormal	12 (11%)	2 (9.1%)	5 (9.6%)	5 (13%)	
Normal	93 (83%)	15 (68%)	45 (87%)	33 (87%)	
Depression	4.5 (3.7)	7.2 (4.3)	4.2 (3.4)	3.4 (2.9)	**<0.001**
Depression category					**0.006**
Abnormal	10 (8.9%)	6 (27%)	3 (5.8%)	1 (2.6%)	
Borderline abnormal	16 (14%)	5 (23%)	8 (15%)	3 (7.9%)	
Normal	86 (77%)	11 (50%)	41 (79%)	34 (89%)	

[1]Mean (SD); n (%)

[2]One-way ANOVA; Fisher's exact test

3 Regression Result

- Attending religious meeting associate with
  1. self-perceived devotion
  2. faith
  3. positive religious coping
  4. depression
  5. intrusion perceived stress
- Personal praying life associate with
  1. self-perceived devotion
  2. peace
  3. faith
  4. positive religious coping
- Studying religious books/material
  - no relationship with any outcomes

- 出席宗教聚會與以下因素相關
  1. 自我認知的虔誠
  2. 信心
  3. 正面的宗教應對方法
  4. 抑鬱 5. 入侵感知壓力
- 個人祈禱生活與以下因素相關
  1. 自我認知的虔誠
  2. 平安
  3. 信心
  4. 正面的宗教應對方法
- 學習宗教書籍/材料
  - 與任何結果無關

Table 7: Rregression on self-perceived devotion

Characteristic	Beta	95% CI[1]	p
(Intercept)	3.19	-1.08, 7.46	0.141
Attending religious meeting			
< 1/week	—	—	
1/week	1.90	1.03, 2.78	**<0.001**
> 1/week	2.29	1.22, 3.36	**<0.001**
Personal pray			
< 1/day	—	—	
>= 1/day	1.11	0.33, 1.90	**0.006**
Studying religious books/materials			
< 1/day	—	—	
>= 1/day	0.00	-0.82, 0.82	0.997
R^2	0.417		

[1]CI = Confidence Interval

Table 8: Regression on spirituality

	Meaning			Peace			Faith		
Characteristic	Beta	95% CI[1]	p	Beta	95% CI[1]	p	Beta	95% CI[1]	p
(Intercept)	9.42	5.06, 13.78	**<0.001**	3.03	-4.91, 10.98	0.450	7.44	-0.60, 15.48	0.069
Attending religious meeting									
< 1/week	—	—		—	—		—	—	
1/week	0.00	-0.90, 0.90	0.998	0.80	-0.83, 2.43	0.334	1.67	0.01, 3.32	**0.048**
> 1/week	0.37	-0.73, 1.47	0.503	1.61	-0.39, 3.61	0.112	2.51	0.49, 4.53	**0.016**
Personal pray									
< 1/day	—	—		—	—		—	—	
>= 1/day	0.61	-0.20, 1.41	0.138	1.57	0.11, 3.03	**0.036**	1.72	0.24, 3.20	**0.023**
Studying religious books/materials									
< 1/day	—	—		—	—		—	—	
>= 1/day	0.19	-0.65, 1.03	0.651	0.24	-1.29, 1.77	0.756	0.14	-1.41, 1.68	0.862
R^2	0.253			0.280			0.370		

[1]CI = Confidence Interval

Table 9: Rregression on religious coping

	Positive			Negative		
Characteristic	Beta	95% CI[1]	p	Beta	95% CI[1]	p
(Intercept)	8.27	-3.34, 19.88	0.160	15.27	9.59, 20.95	**<0.001**
Attending religious meeting						
< 1/week	—	—		—	—	
1/week	2.54	0.16, 4.93	**0.037**	-0.09	-1.26, 1.08	0.881
> 1/week	3.46	0.54, 6.37	**0.021**	-0.02	-1.44, 1.41	0.980
Personal pray						
< 1/day	—	—		—	—	
>= 1/day	3.83	1.69, 5.96	**<0.001**	0.36	-0.69, 1.41	0.497
Studying religious books/materials						
< 1/day	—	—		—	—	
>= 1/day	1.33	-0.90, 3.56	0.238	-0.63	-1.72, 0.46	0.251
R^2	0.397			0.242		

[1]CI = Confidence Interval

Table 10: Regression on mental health

	Anxiety			Depression		
Characteristic	Beta	95% CI[1]	p	Beta	95% CI[1]	p
(Intercept)	14.08	5.68, 22.47	**0.001**	8.33	-0.44, 17.09	0.062
Attending religious meeting						
< 1/week	—	—		—	—	
1/week	-1.05	-2.77, 0.68	0.232	-2.34	-4.14, -0.54	**0.011**
> 1/week	-0.78	-2.89, 1.32	0.461	-3.10	-5.30, -0.90	**0.006**
Personal pray						
< 1/day	—	—		—	—	
>= 1/day	-0.89	-2.44, 0.65	0.254	-1.54	-3.16, 0.07	0.061
Studying religious books/materials						
< 1/day	—	—		—	—	
>= 1/day	-1.19	-2.80, 0.42	0.146	0.17	-1.51, 1.86	0.838
R^2	0.314			0.347		

[1]CI = Confidence Interval

Table 11: Regression on perceived stress

	Intrusion			Avoidance			Hyperarousal		
Characteristic	Beta	95% CI[1]	p	Beta	95% CI[1]	p	Beta	95% CI[1]	p
(Intercept)	13.88	3.88, 23.88	**0.007**	20.70	7.85, 33.54	**0.002**	13.21	6.39, 20.04	**<0.001**
Attending religious meeting									
< 1/week	—	—		—	—		—	—	
1/week	-2.77	-4.83, -0.71	**0.009**	-1.99	-4.63, 0.65	0.138	-0.99	-2.39, 0.41	0.164
> 1/week	-3.08	-5.59, -0.57	**0.017**	-2.03	-5.26, 1.19	0.214	-0.98	-2.69, 0.74	0.260
Personal pray									
< 1/day	—	—		—	—		—	—	
>= 1/day	1.12	-0.73, 2.96	0.232	1.62	-0.75, 3.98	0.178	0.25	-1.01, 1.51	0.695
Studying religious books/materials									
< 1/day	—	—		—	—		—	—	
>= 1/day	-0.24	-2.16, 1.68	0.806	0.40	-2.07, 2.87	0.748	-0.26	-1.57, 1.05	0.693
R^2	0.259			0.300			0.415		

[1]CI = Confidence Interval

4 Conclusion

80% of the respondents had attended the religious meeting at least every week and 60% had prayed at least every day. Although people attending more religious meeting every week tends to pray more every week, this relationship is far from our expectation. Besides, 6.2% and 8.9% were abnormal for anxiety and depression respectively. Our findings suggested that religious behavior substantially affected individuals' spirituality, perceived, religious coping, and mental health. Furthermore, studying religious books/materials did not affect the above outcome. The finding also begs the question of what factors are which drive the phenomenon of studying. One key policy implication of the present study is that the church (clergy) should emphasize the importance of praying to enhance their mental health and spirituality.

80% 的參加者至少每週都出席宗教聚會，並且 60% 至少每天都祈禱。雖然每週井席更多宗教聚會的人往往每週都祈禱更多，但這種關係與我們的期望相距甚遠。此外，分別有 6.2% 和 8.9% 的參加者有焦慮和抑鬱。我們的研究結果表明，宗教行為極大地影響了個人的靈性、自我認知壓力、宗教應對和心理健康。此外，閱讀宗教書籍不會影響上述結果。這一發現促使我們提出疑問，是哪些因素導致這現象出現。本研究的一個關鍵政策意義，便是教會（傳道同工）應該更強調祈禱的重要性，以增強教友的心理健康和靈性的發展。

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6 Appendix

- FACIT-Sp-12
- Meaning

  2. 我有活下去的理由
  3. 我的生命一直很豐盛
  4. 我覺得我的生命有目標
  5. 我的生命缺乏意義和目標

- Peace

  1. 我感到平靜
  2. 我很難感受到內心的平靜
  3. 我能夠從我的內心深處找到安適
  4. 我感受到自己身心和諧

- Faith

  9. 我從我的信念或信仰中找到安適
  10. 我從我的信念或信仰中找到力量
  11. 我的疾病使我的信念或信仰更加堅定
  12. 我知道不論我的疾病如何變化，一切都將安好

- IES-R
- Intrusion

  1. 任何有關那件事的事物也會引起我對那件事的感覺
  2. 我難於保持安睡
  3. 其他事情不停地令我想起那件事
  4. 我會無意中想起那件事
  5. 關於那件事的影像在我腦海中突然浮現出來
  6. 我覺得我的行動或者感覺好像回復到那件事發生時的行動和感覺一樣
  7. 我對那件事的強烈感覺一陣一陣地湧現出來
  8. 我夢到那件事

- Avoidance

  5. 每當我想起那件事或者有其他事物令我想起那件事時，我會盡力避免讓自己心煩意亂
  6. 我覺得那件事好像從未發生過或並非真實
  7. 我避開一些會令我回憶起那件事的事物
  8. 我嘗試不想起那件事
  9. 我察覺到我仍然對那件事有很多感受，但我沒有去處理它們
  10. 我對那件事的感覺是有點兒麻木
  11. 我嘗試忘記那件事
  12. 我嘗試不談論有關那件事的話題

- Hyperarousal

  4. 我覺得煩躁和忿怒
  5. 我神經過敏及容易被嚇得跳起來
  6. 我難於入睡
  7. 我難於集中精神
  8. 令我想起那件事的事物使我身體產生反應，例如：流汗、呼吸困難、作嘔、或者強烈心跳
  9. 我覺得自己警覺性很高，處處提防

- Brief RCOPE
- Positive

  1. 我尋求更加親近上帝。
  2. 我尋求上帝的愛與看顧。
  3. 我尋求上帝幫助我化解我的怒氣
  4. 我試著跟上帝一起實行我的計畫
  5. 我試著了解上帝在這樣的處境下會如何堅固我
  6. 我求上帝赦免我的罪

7. 我會藉著定睛在信仰上使我停止憂慮

- Negative

8. 我懷疑上帝是否已經棄我於不顧
9. 我覺得上帝懲罰我是因為我不夠愛主
10. 我懷疑是不是我做了甚麼導致上帝懲罰我。
11. 我懷疑上帝對我的愛
12. 我懷疑我的教會是否已經棄我於不顧
13. 我認定這一切困難是出於魔鬼
14. 我質疑上帝的能力